

PHASE 1 — INTERNET & WEB BASICS

1. Introduction to Web Development

- What is Web Development
 - Frontend vs Backend vs Full Stack
 - Websites vs Web Apps
 - Static vs Dynamic websites
 - Client-Server architecture
 - Browser working
 - Request-Response cycle
 - HTTP & HTTPS
 - Domain name
 - DNS
 - Hosting basics
 - Rendering process
 - Developer Roadmap
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PHASE 2 — HTML COMPLETE

2. HTML Basics

- What is HTML
- Boilerplate structure
- Tags & Elements
- Attributes
- Head & Body
- VS Code setup
- Live Server

3. Text Formatting Tags

- Heading tags
- Paragraph
- Bold
- Italic
- Underline

- Superscript/Subscript
- HR tag
- BR tag
- Pre tag

4. Lists

- Ordered list
- Unordered list
- Nested lists
- Description lists

5. Anchor Tags

- Internal links
- External links
- Relative paths
- Absolute paths
- Opening in new tab

6. Images & Multimedia

- Image tag
- Alt attribute
- Width & Height
- Audio tag
- Video tag
- iframe

7. Tables

- Table structure
- tr, td, th
- thead, tbody
- colspan
- rowspan

8. Forms Complete

- Form tag
- Input types

- Labels
- Placeholder
- Name/value
- Checkbox
- Radio buttons
- Select dropdown
- Textarea
- Buttons
- Validation attributes
- Required field
- Pattern
- Email validation

9. Semantic HTML

- Semantic tags
- Header
- Main
- Footer
- Section
- Article
- Aside
- Nav
- Accessibility basics

10. HTML5 Advanced

- Semantic improvements
- Audio/video APIs
- Forms validation
- Meta tags
- SEO basics
- favicon

11. Introduction to CSS

- Inline CSS
- Internal CSS
- External CSS
- Syntax
- Selectors

12. CSS Selectors

- Universal selector
- Element selector
- Class selector
- ID selector
- Grouping selector
- Descendant selector
- Child selector
- Adjacent sibling
- Attribute selector
- Pseudo classes
- Pseudo elements

13. Colors & Units

- RGB
- HEX
- HSL
- px
- rem
- em
- vh/vw
- Percentages

14. Text & Font Styling

- font-family
- font-size
- font-weight

- line-height
- text-align
- text-decoration
- letter-spacing
- Google Fonts

15. Box Model

- Margin
- Padding
- Border
- Width & Height
- box-sizing

16. Display Property

- block
- inline
- inline-block
- none

17. Positioning

- static
- relative
- absolute
- fixed
- sticky
- z-index

18. Backgrounds

- background-color
- background-image
- gradients
- cover/contain

19. Flexbox COMPLETE

- Main axis
- Cross axis

- justify-content
- align-items
- flex-wrap
- gap
- align-self
- flex-grow
- flex-shrink

20. CSS Grid

- Grid container
- Grid items
- Columns & rows
- fr unit
- Grid areas
- Auto-fit/Auto-fill

21. Transitions & Transform

- transition
- transform
- rotate
- scale
- skew
- translate

22. Animations

- @keyframes
- animation properties
- Infinite animations

23. Responsive Design

- Media queries
 - Mobile-first design
 - Breakpoints
 - Responsive images
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PHASE 4 — BOOTSTRAP

24. Bootstrap Basics

- Bootstrap CDN
- Containers
- Grid system
- Rows & columns

25. Bootstrap Components

- Navbar
- Cards
- Buttons
- Alerts
- Forms
- Carousel
- Modal
- Accordion

26. Bootstrap Utilities

- Margin/padding utilities
 - Flex utilities
 - Text utilities
 - Responsive utilities
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PHASE 5 — JAVASCRIPT COMPLETE

28. Introduction to JavaScript

- What is JavaScript
- How JS works
- Interpreted language
- JS engine
- Linking JS with HTML

29. Variables & Data Types

- var

- let
- const
- Primitive types
- Non-primitive types
- typeof

30. Operators

- Arithmetic
- Assignment
- Comparison
- Logical
- Ternary
- Increment/decrement

31. Conditional Statements

- if
- else
- else-if
- switch

32. Loops

- for loop
- while
- do while
- Nested loops
- break & continue

33. Strings

- String methods
- Template literals
- Escape characters

34. Arrays

- Array methods
- push/pop
- shift/unshift

- splice/slice
- map
- filter
- reduce
- forEach

35. Objects

- Object creation
- Key-value pairs
- Nested objects
- Object methods
- this keyword

36. Functions

- Function declaration
- Function expression
- Arrow functions
- Callback functions
- Higher-order functions
- Scope
- Hoisting

37. Advanced JavaScript Concepts

- Execution context
- Call stack
- Lexical scope
- Closures
- Hoisting
- Memory allocation
- Garbage collection

38. DOM Manipulation COMPLETE

- DOM tree
- Selecting elements
- querySelector

- getElementById
- innerText
- innerHTML
- textContent
- Changing styles
- Creating elements
- Removing elements

39. Events

- onclick
- addEventListener
- Mouse events
- Keyboard events
- Form events
- Event object

40. Event Bubbling & Delegation

- Capturing phase
- Bubbling phase
- stopPropagation
- Event delegation

41. Browser APIs

- setTimeout
- setInterval
- Local Storage
- Session Storage

42. Object Oriented Programming (OOP)

- Classes
- Objects
- Constructors
- Inheritance
- Encapsulation
- Prototypes

43. ES6+ Features

- Destructuring
- Spread operator
- Rest operator
- Default parameters
- Optional chaining
- Modules
- Import/export

44. Async JavaScript

- Synchronous vs asynchronous
- Callback hell
- Promises
- Promise chaining
- async/await

45. APIs & AJAX

- Fetch API
- Axios
- JSON
- HTTP methods
- REST APIs
- API calls
- Handling responses
- Error handling

46. Mini Projects in JavaScript

Usually:

- Todo App
- Simon Game
- Weather App
- Calculator
- Notes App
- Spotify Clone basics